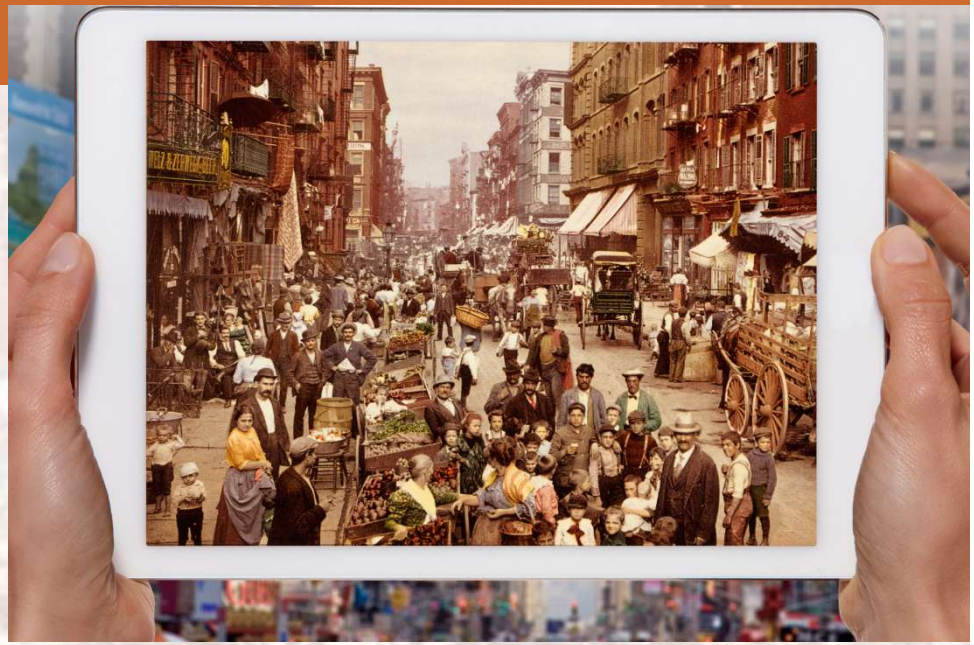
經濟學



CHAPTER

2

Thinking Like an Economist

National Central University

Look for the answers to these questions:

- What are economists' two roles? How do they differ?
- What are models? How do economists use them?
- What are the elements of the Circular-Flow Diagram (循環流向圖)? What concepts does the diagram illustrate?
- How is the Production Possibilities Frontier (生產可能線) related to opportunity cost? What other concepts does it illustrate?
- What is the difference between microeconomics and macroeconomics? Between positive (實證性) and normative (規範性)?



The Economist as a Scientist

- Economists play two roles:
  1. Scientists: try to explain the world
  2. Policy advisors: try to improve it
- As scientists, economists employ the scientific method
  - Dispassionate development and testing of theories about how the world works

大膽假設,小心求證



The Economist as a Scientist

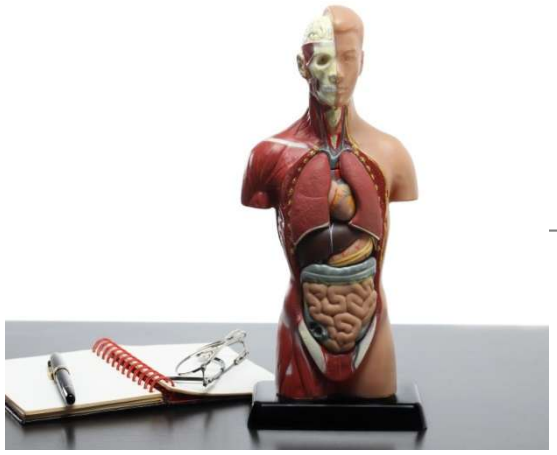
- Assumptions 看清重要的事實
  - Simplify the complex world and make it easier to understand
  - Example: to study international trade, assume two countries and two goods
- Economists use models to study economic issues
  - Highly simplified representation of a more complicated reality



The Economist as a Scientist

- Examples of models:
  - The model teeth at the dentist's office
  - A model of human anatomy from high school biology class
  - A road map







©ittipon/Shutterstock.com

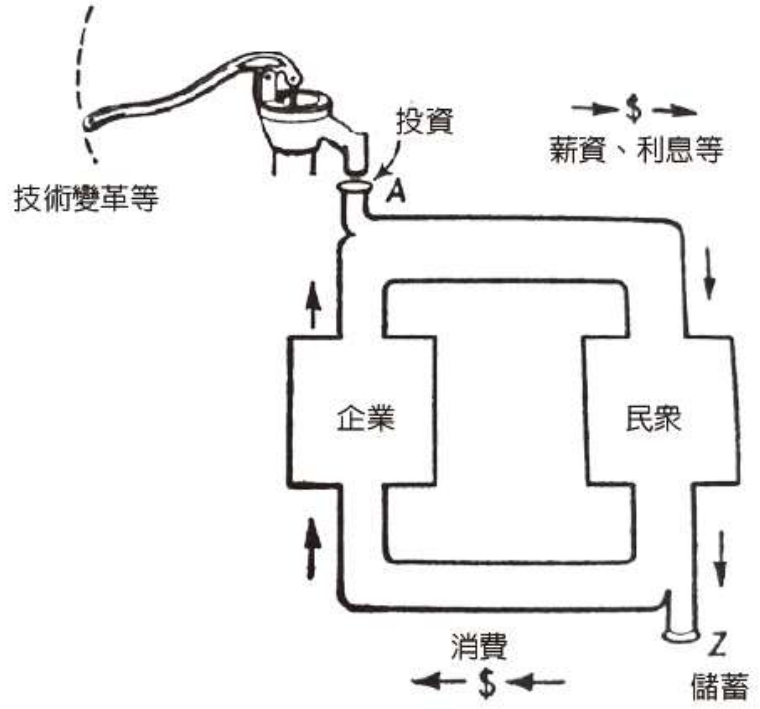


The Economist as a Scientist

- Circular-flow diagram (循環流向圖)
  - Visual model of the economy
  - Shows how dollars flow through markets among households and firms
- Two decision makers
  - Firms and Households
- Interacting in two markets
  - Market for goods and services
  - Market for factors of production (inputs)

Circular-flow diagram

麻省理工學院經濟學系教授保羅．薩繆森，為了教導理工科學生讀懂經濟學理論，他把所有總體經濟學理論圖表化，最有名的圖示是「循環流向圖」，這張圖示用水在水管系統當中的流動說明了所得在經濟體中如何不斷循環流轉，於是理工科的這些門外漢一看就懂。



薩繆森1948年的循環流向圖。
該圖描繪所得在經濟體中的流轉，就像是水在水管系統當中流動一樣。

Figure 1 The circular flow

Households:

- Own the factors of production, sell/rent them to firms for income
- Buy and consume goods & services

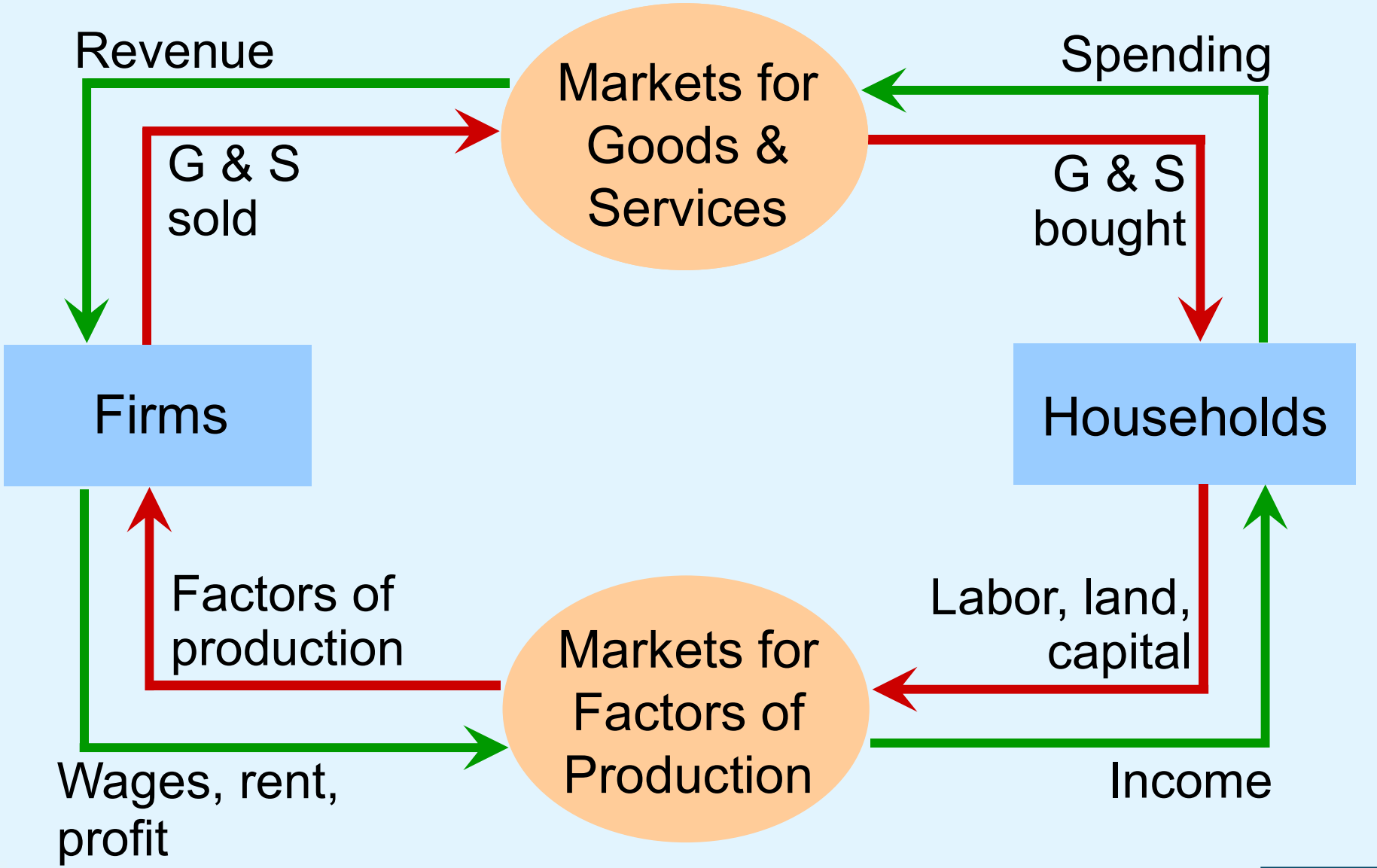
Firms

Households

Firms:

- Buy/hire factors of production, use them to produce goods and services
- Sell goods & services

Figure 1 The circular flow



Question

- Draw a circular-flow diagram.
- Identify the parts of the model that correspond to the flow of goods and services and the flow of dollars for each of the following activities
  - (A)Selena pays a storekeeper $1 for a quart of milk
  - (B)Stuart earns $8 per hour working at a fast-food restaurant
  - (C)Shanna spends $40 to get a haircut
  - (D)Salma earns $20,000 from her 10 percent owner-ship of Acme Industrial



The PPF

- Production possibilities frontier (生產可能線)
  - A graph: combinations of output that the economy can possibly produce
  - Given the available
    - Factors of production and technology
  - Example:
    - Two goods: computers and wheat
    - One resource: labor (measured in hours)
    - Economy has 50,000 labor hours per month available for production

PPF Example

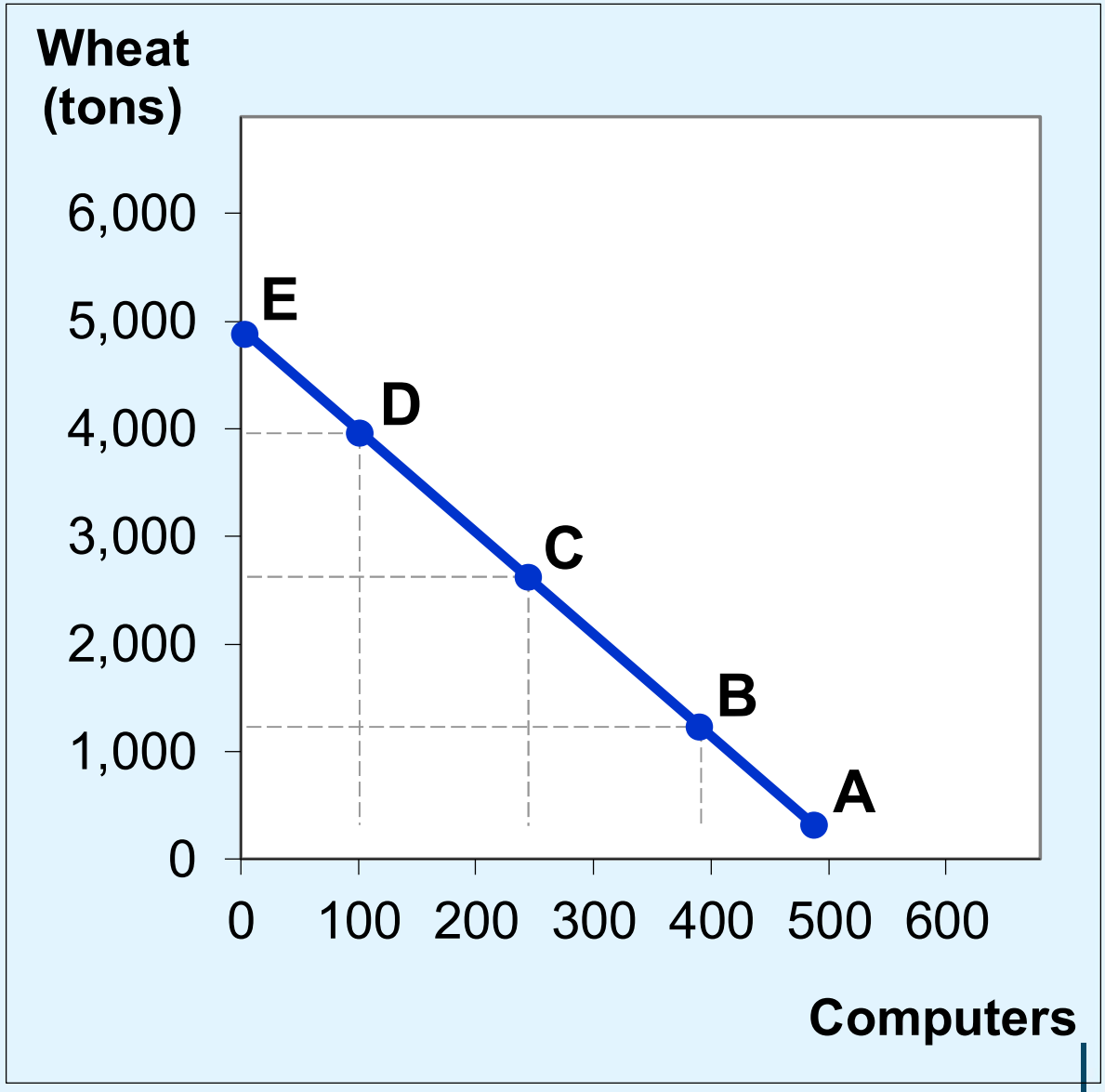
Producing one computer requires 100 hours labor.

Producing one ton of wheat requires 10 hours labor.

	Employment of labor hours		Production	
	Computers	Wheat	Computers	Wheat
A	50,000	0	500	0
B	40,000	10,000	400	1,000
C	25,000	25,000	250	2,500
D	10,000	40,000	100	4,000
E	0	50,000	0	5,000

PPF Example

Point on graph	Production	
	Com-puters	Wheat
A	500	0
B	400	1,000
C	250	2,500
D	100	4,000
E	0	5,000



Active Learning 1 — Points off the PPF

On the graph above, find the point that represents (100 computers, 3000 tons of wheat), label it **F**.

- Would it be possible for the economy to produce this combination of the two goods? Why or why not?

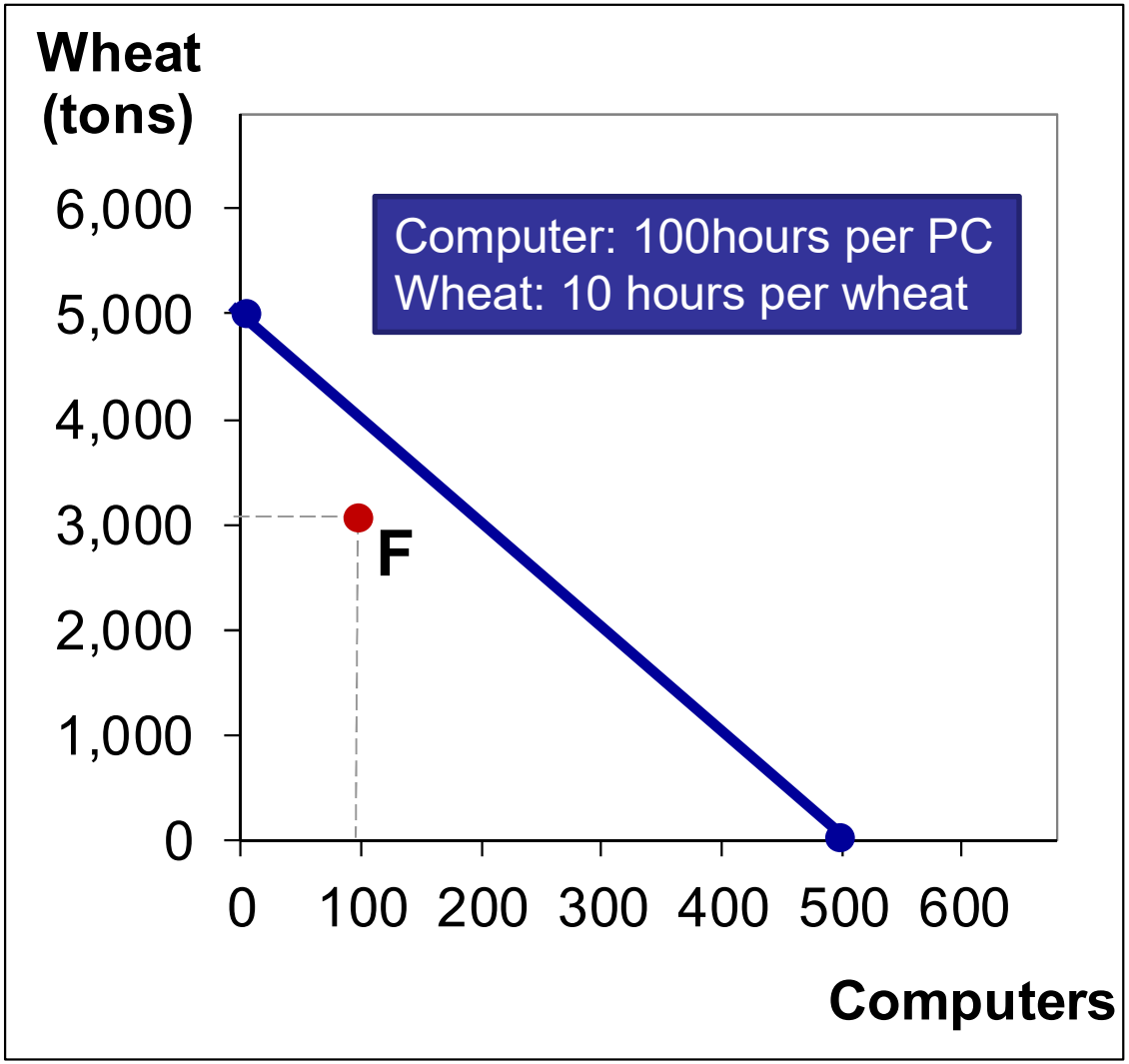
Next, find the point that represents (300 computers, 3500 tons of wheat), label it **G**.

- Would it be possible for the economy to produce this combination of the two goods?

Active Learning 1

Answers

- Point F: 100 computers, 3000 tons wheat
- Requires 40,000 hours of labor
- Possible but not efficient: could get more of either good without sacrificing any of the other

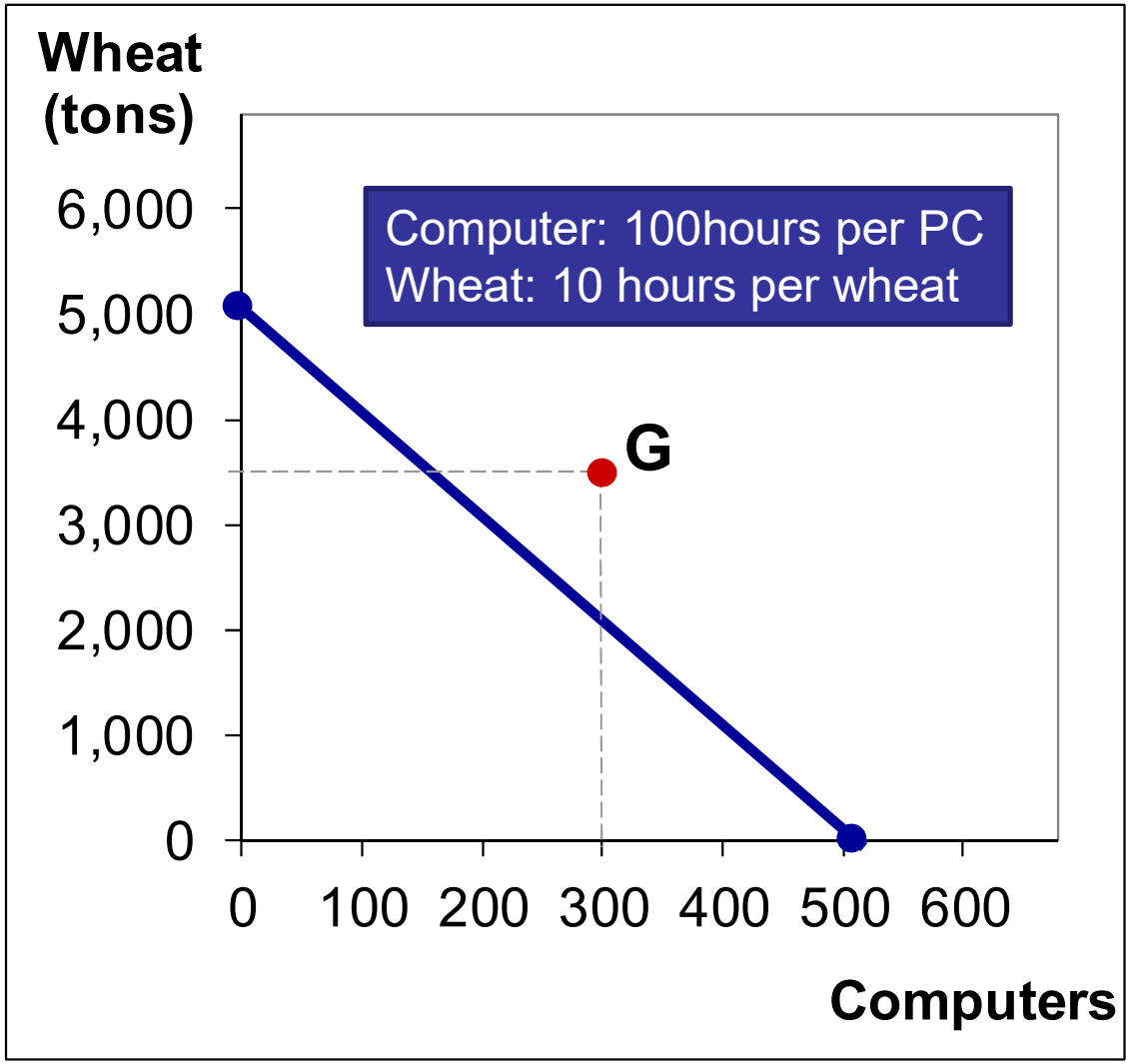


100x100+3000x10=40,000<50,000

Active Learning 1

Answers

- Point G: 300 computers, 3500 tons wheat
- Requires 65,000 hours of labor.
- Not possible because the economy only has 50,000 hours



300x100+3500x10=65,000>50,000



The PPF: What We Know So Far

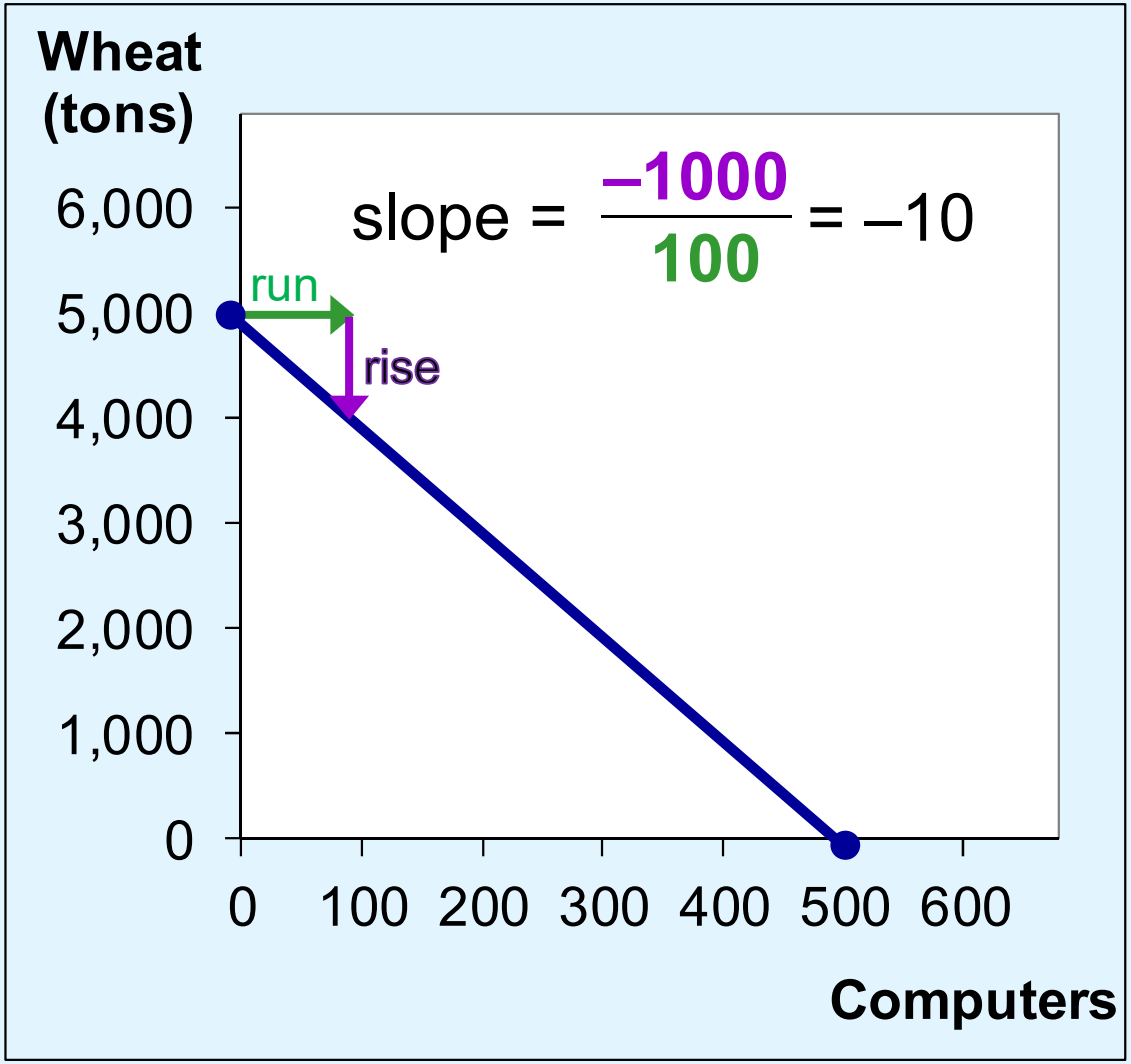
- Points on the PPF (like A – E): possible
  - Efficient: all resources are fully utilized
- Points under the PPF (like F): possible
  - Not efficient: some resources are underutilized (e.g., workers unemployed, factories idle) 未充分使用資源
- Points above the PPF (like G)
  - Not possible



The PPF

- Moving along a PPF
  - Involves shifting resources from the production of one good to the other
- Society faces a tradeoff
  - Getting more of one good requires sacrificing some of the other
- The slope of the PPF
  - The opportunity cost of one good in terms of the other

The PPF and Opportunity Cost

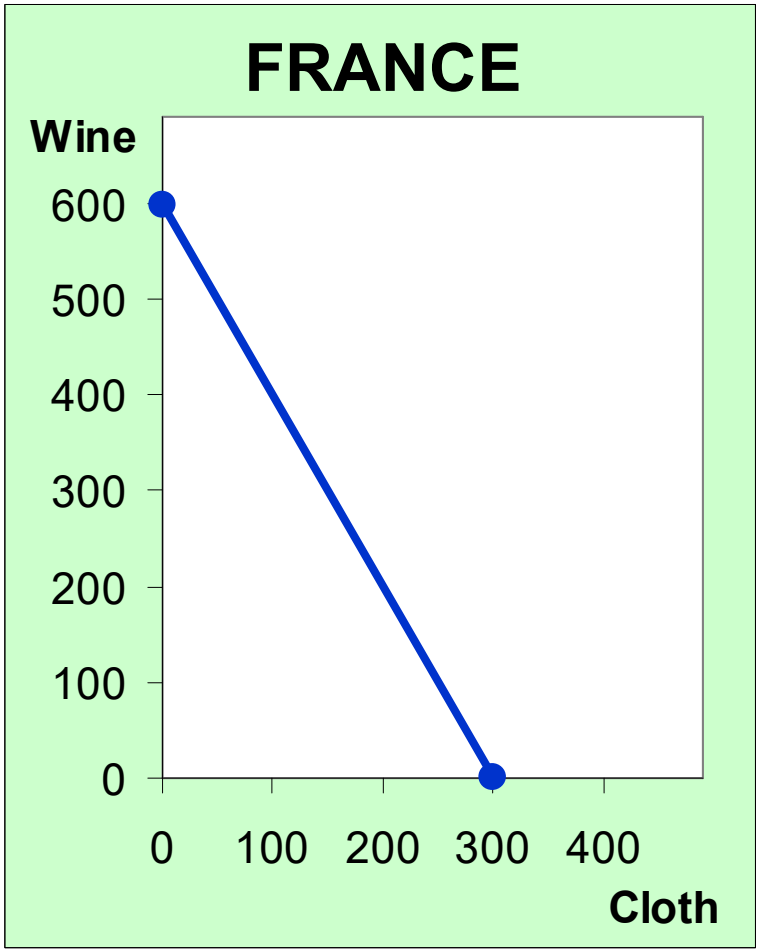


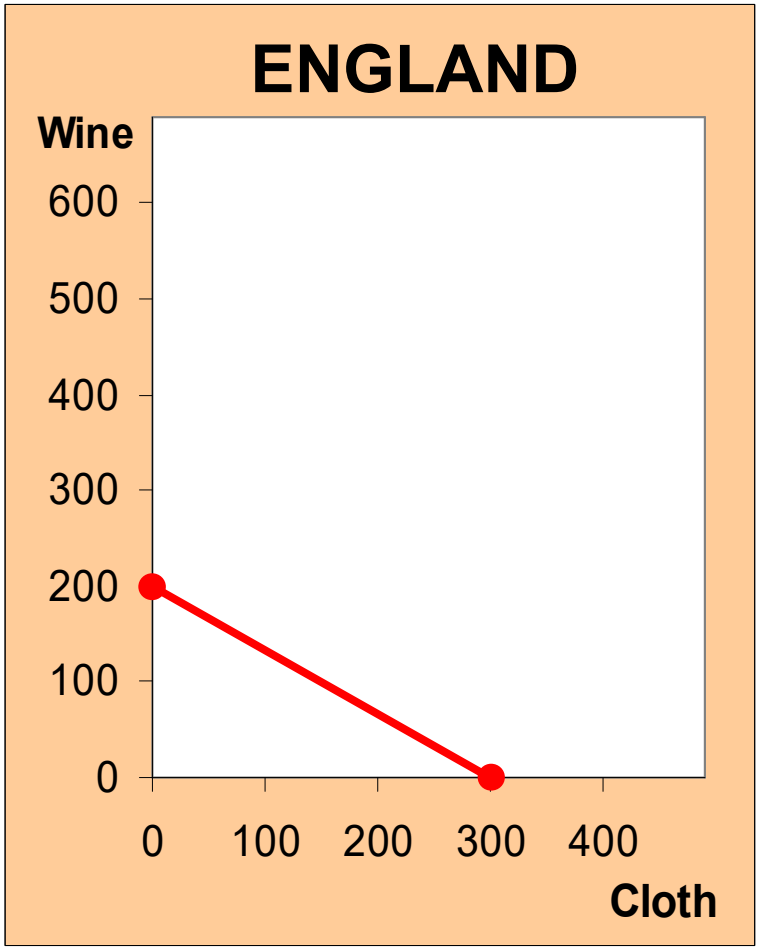
The slope of a line equals the "**rise** over the **run.**"

Opportunity cost of 1 computer = 10 tons of wheat.

Active Learning 2 PPF and Opportunity Cost

In which country is the opportunity cost of cloth lower?

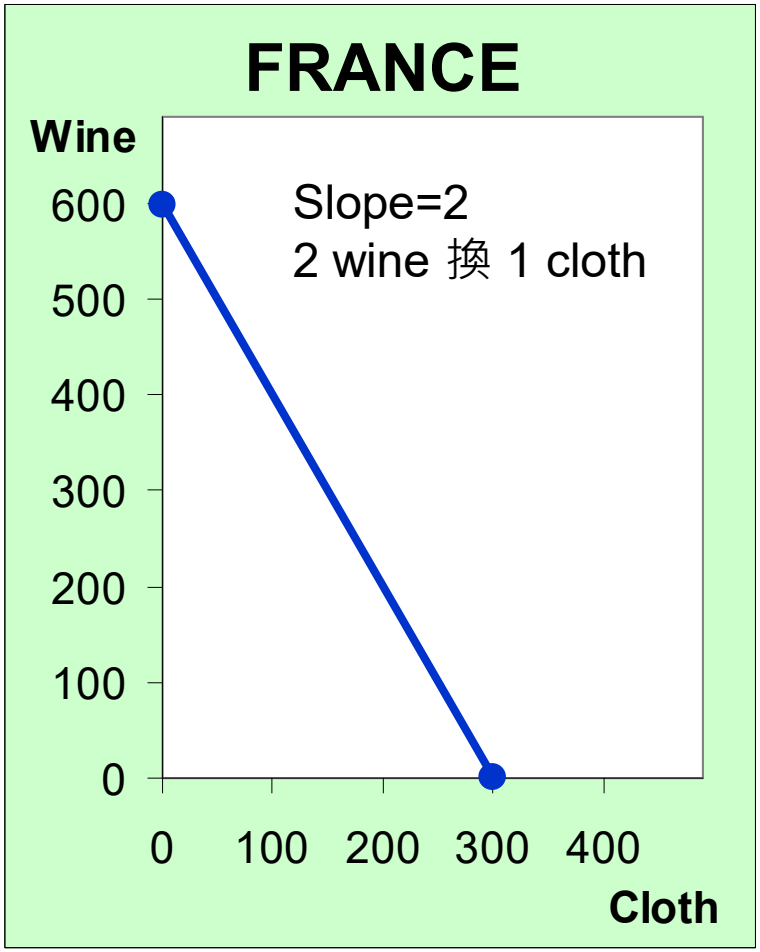


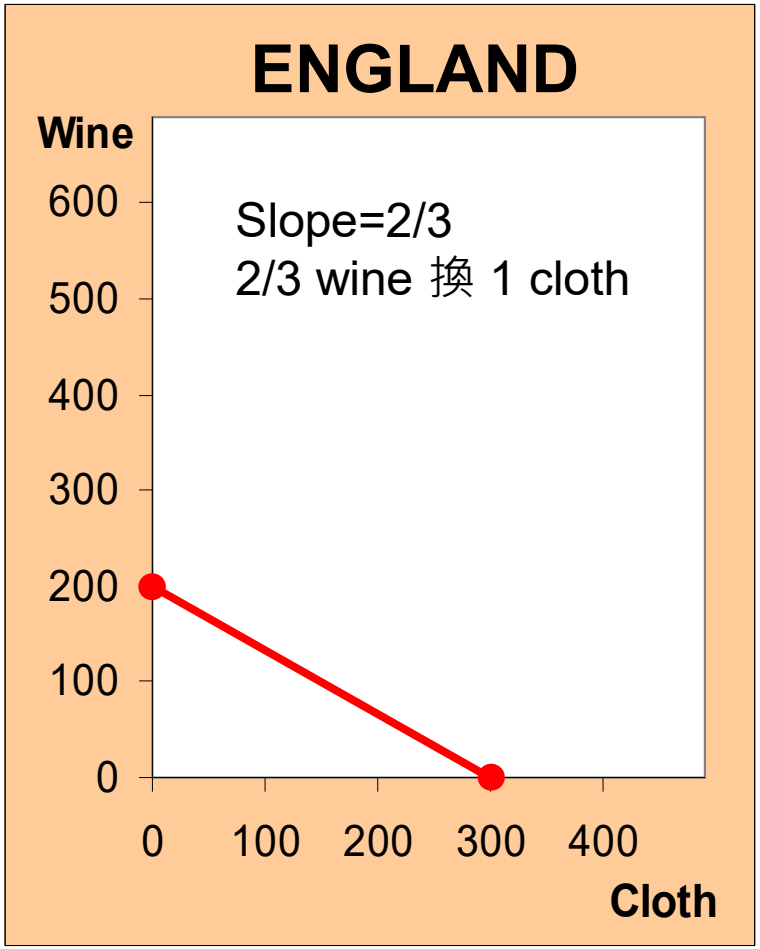


Active Learning 2

Answers

England, because its PPF is not as steep as France's

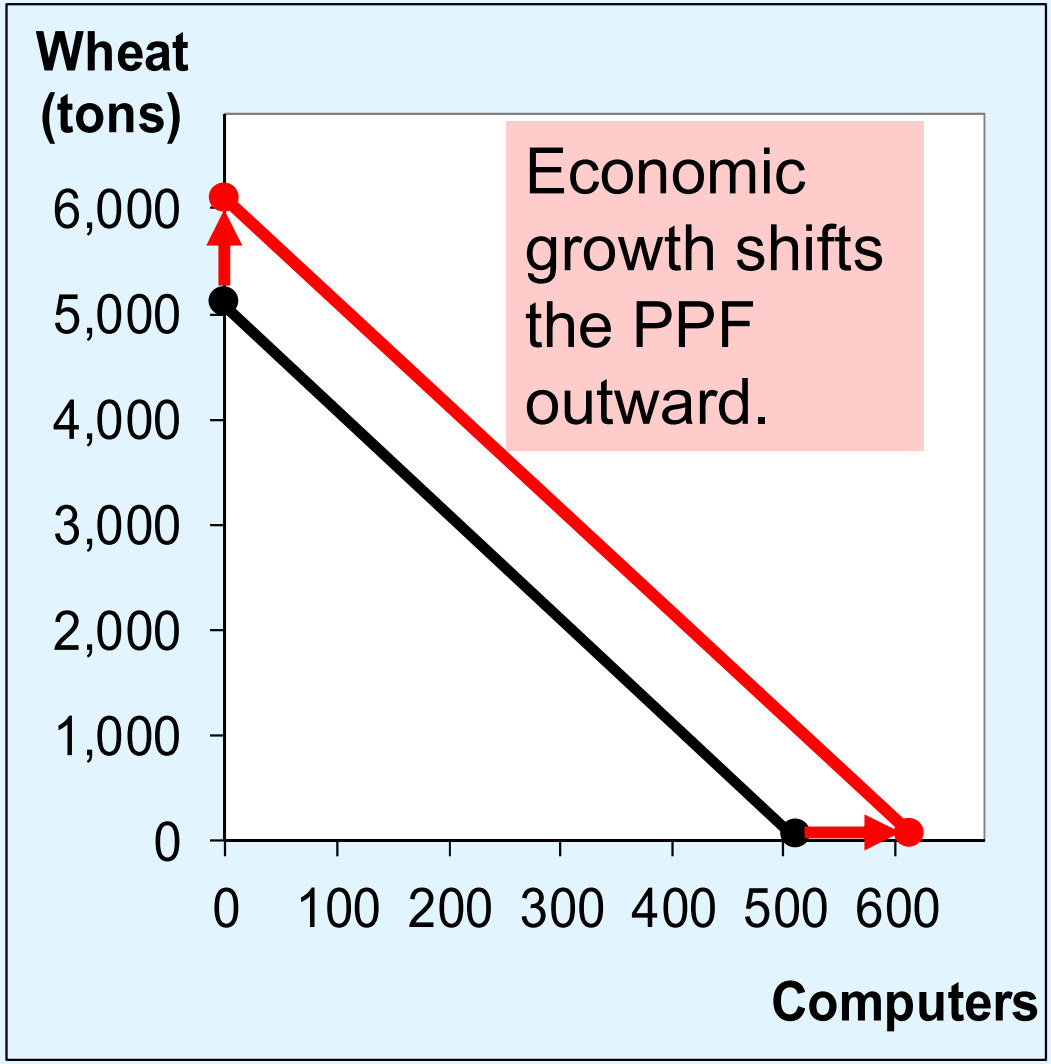




PPF的斜率就是水平軸商品的機會成本

Economic Growth and the PPF

With additional resources or an improvement in technology, the economy can produce more computers, more wheat, or any combination in between.

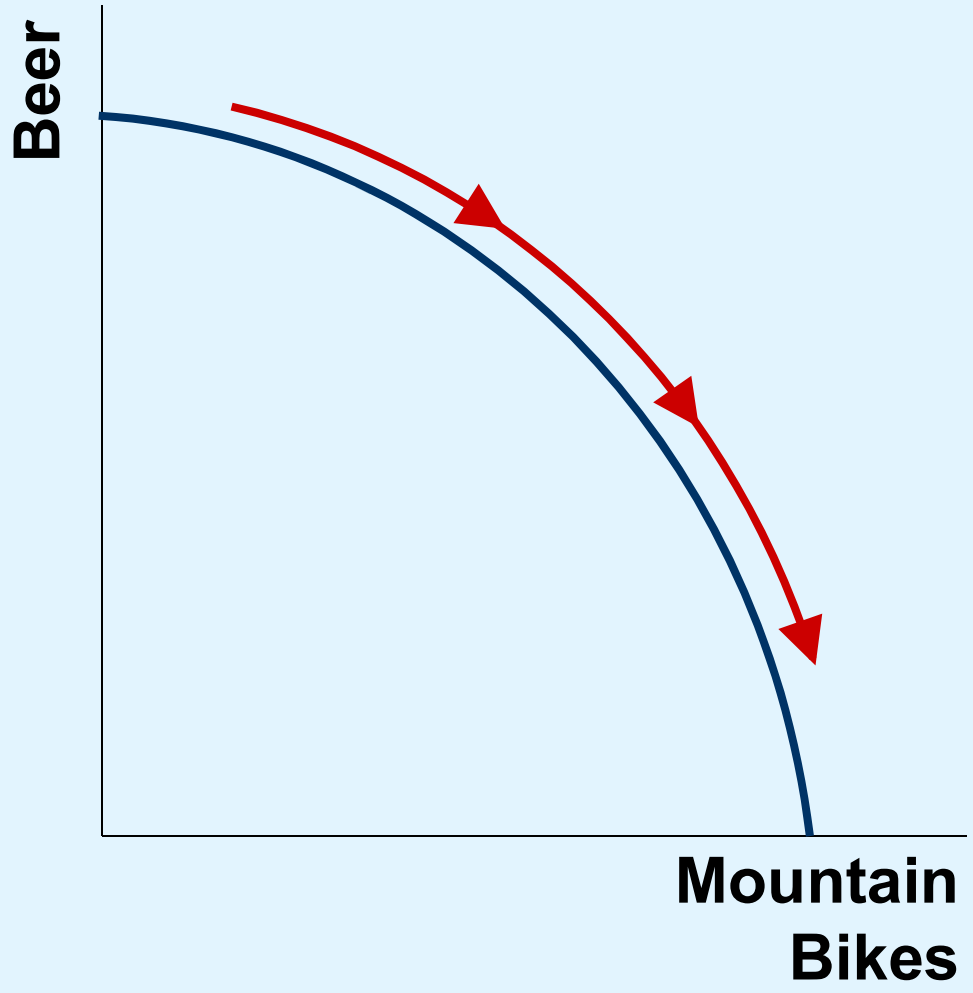




The Shape of the PPF

- Shape of the PPF
  - Straight line: constant opportunity cost
    - Previous example: the opportunity cost of 1 computer is 10 tons of wheat
  - Bowed outward: increasing opportunity cost
    - As more units of a good are produced, we need to give up increasing amounts of the other good produced

Why the PPF Might Be Bowed Outward

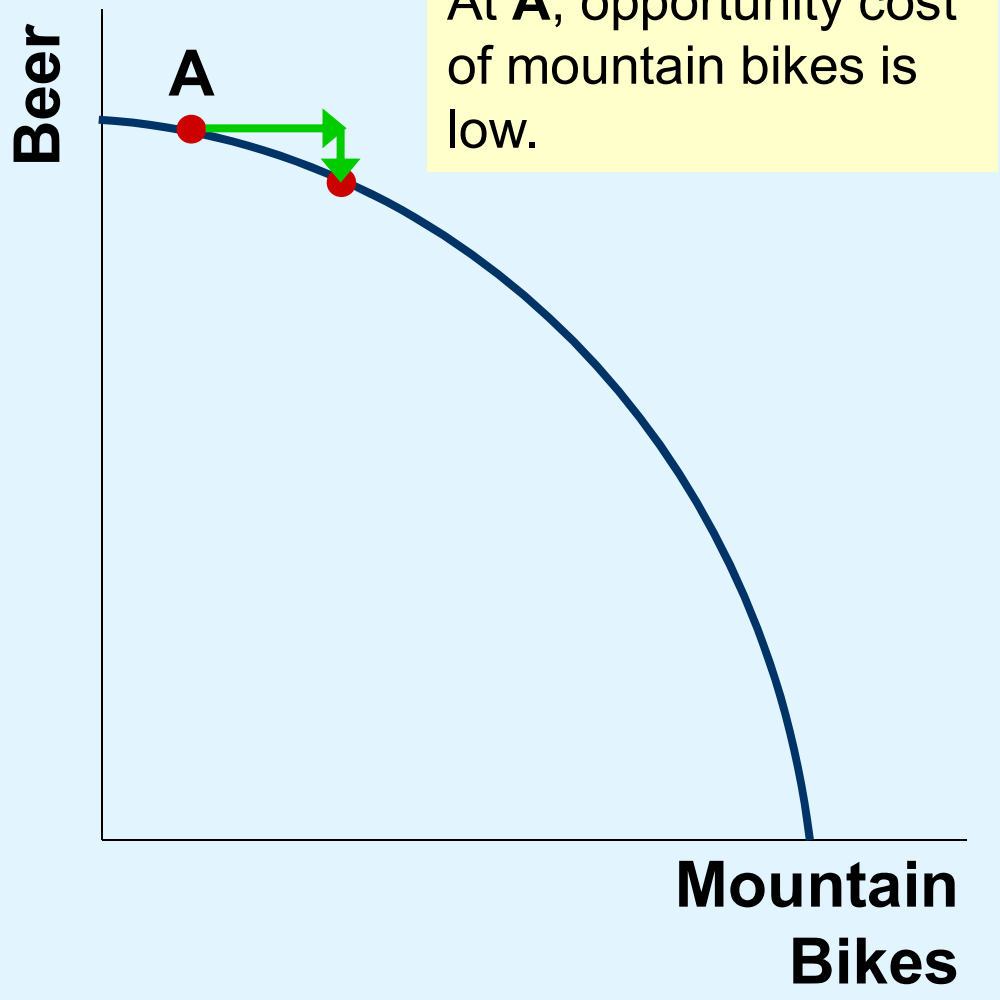


As the economy shifts resources from beer to mountain bikes:

PPF becomes steeper

and the opportunity cost of mountain bikes increases.

Why the PPF Might Be Bowed Outward



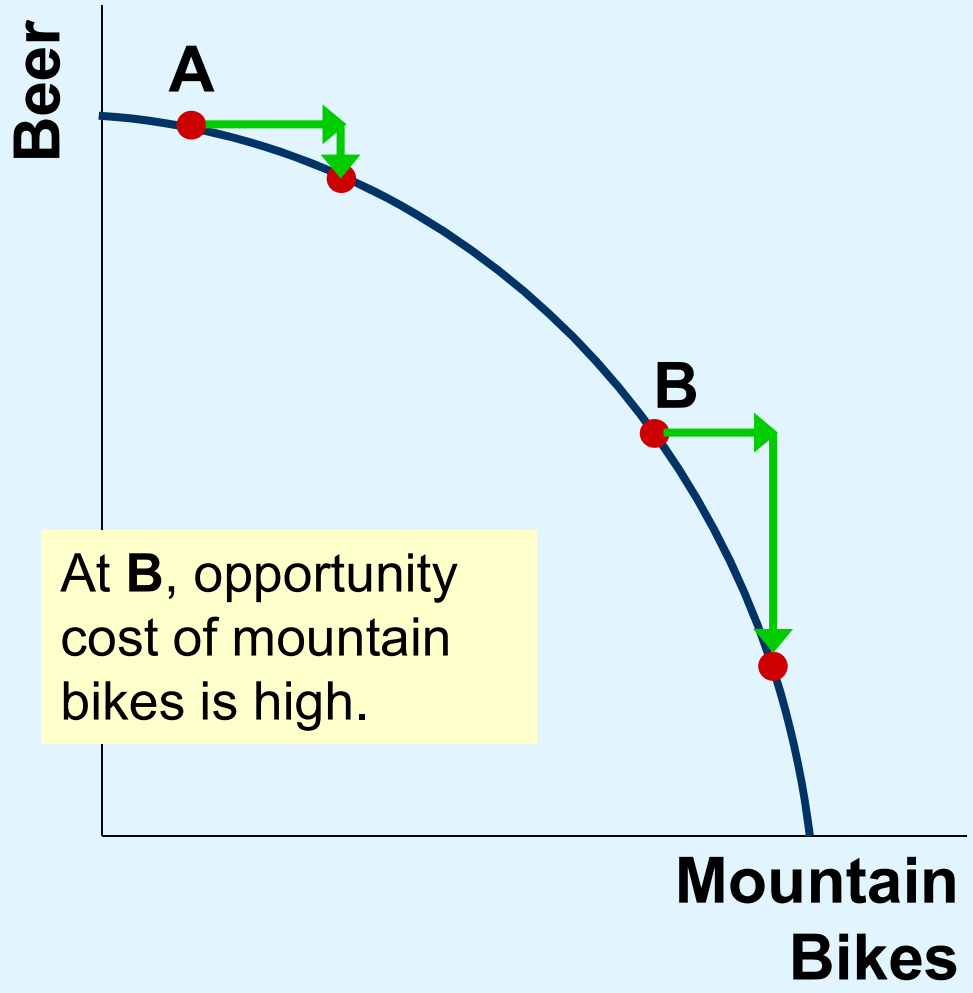
At **A**, opportunity cost of mountain bikes is low.

At point A, most workers are producing beer, even those who are better suited to building bikes.

So, do not have to give up much beer to get more bikes.

啤酒廠人很多,裡面總是有幾位釀酒技能普通的技師,甚至有人專精製造腳踏車
分幾個人到登山車廠不會掉太多產能

Why the PPF Might Be Bowed Outward



At B, most workers are producing bikes. The few left in beer production are the best brewers.

Producing more bikes would require shifting some of the best brewers away from beer production causing a big drop in beer output.

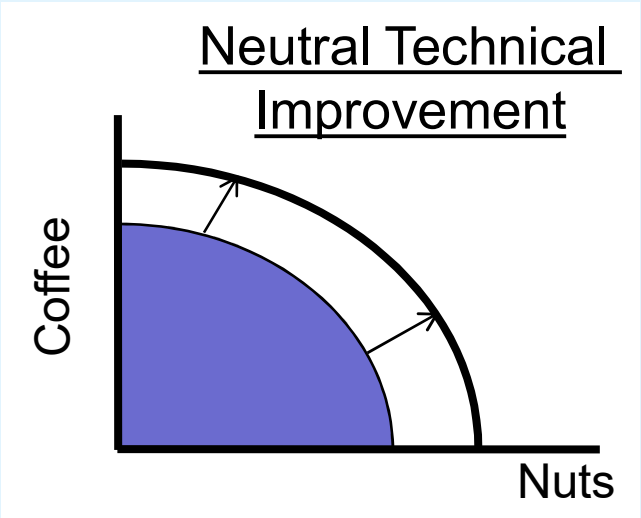
啤酒廠人很少時,個個都是菁英;分幾個人到登山車廠時,產能掉得比較快

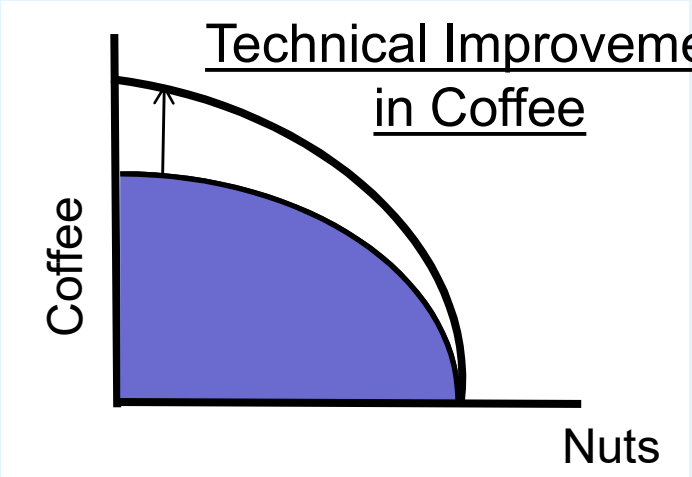


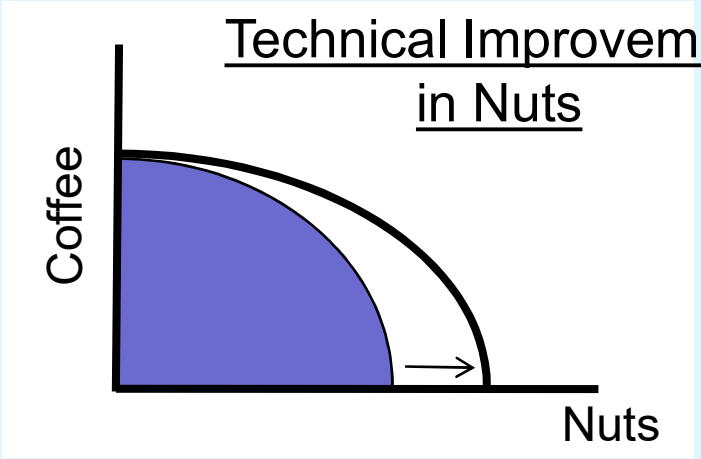
Why the PPF Might Be Bowed Outward

- The PPF is bowed outward when:
  - Different opportunity costs of producing one good in terms of the other
    - Different workers have different skills
    - There is some other resource, or mix of resources with varying opportunity costs
      - E.g., different types of land suited for different uses

Shifts in PPF (生產可能線)









Question

- Use a production possibilities frontier to illustrate society's trade-off between two "goods" – a clean environment and the quantity of industrial output.
- Show what happens to the frontier if engineers develop a new way of producing electricity that emits fewer pollutants



Question

- An economy consists of three workers: Larry, Moe, and Curly.
- Each works **10** hours a day and can produce two services: mowing lawns and washing cars.
- In a hour
  - Larry can either mow **1** lawn or wash **1** car
  - Moe can either mow **1** lawn or wash **2** cars
  - Curly can either mow **2** lawns or wash **1** car

X軸割草
Y軸洗車

- Graph the production possibilities frontier for this economy
  - All three spend all their time mowing lawns (A)
  - All three spend all their time washing cars (B)
  - All three spend half their time on each activity (C)
  - Larry spends half his time on each activity, while Moe only washes cars and Curly only mows lawns (D)

Which one is inefficient? Why?



The Economist as a Scientist

- Microeconomics
  - The study of how households and firms make decisions and how they interact in markets
- Macroeconomics
  - The study of economy-wide phenomena, including inflation, unemployment, and economic growth



Question

- Classify the following topics as relating to microeconomics or macroeconomics
  - A. a family's decision about how much income to save
  - B. the effect of government regulations on auto emissions
  - C. the impact of higher national saving on economic growth
  - D. a firm's decision about how many workers to hire
  - E. the relationship between the inflation rate and changes in the quantity of money



The Economist as Policy Adviser

實證性陳述

- Positive statements: descriptive
  - Attempt to describe the world as it is
  - Confirm or refute by examining evidence: “Minimum-wage laws cause unemployment”

規範性陳述

- Normative statements: prescriptive
  - Attempt to prescribe how the world should be: “The government should raise the minimum wage”

Active Learning 3 Positive vs. Normative

Which of these statements are “positive” and which are “normative”? How can you tell the difference?

a. Prices rise when the government increases the quantity of money.
b. The government should print less money.
c. A tax cut is needed to stimulate the economy.
d. An increase in the price of burritos will cause an increase in consumer demand for music downloads.



Active Learning 3 Answers

a. Prices rise when the government increases the quantity of money.

Positive – describes a relationship, could use data to confirm or refute.

b. The government should print less money.

Normative – this is a value judgment, cannot be confirmed or refuted.

Active Learning 3 Answers

c. A tax cut is needed to stimulate the economy.

Normative – another value judgment.

d. An increase in the price of burritos will cause an increase in consumer demand for music downloads

Positive – describes a relationship.

Note that a statement need not be true to be positive.



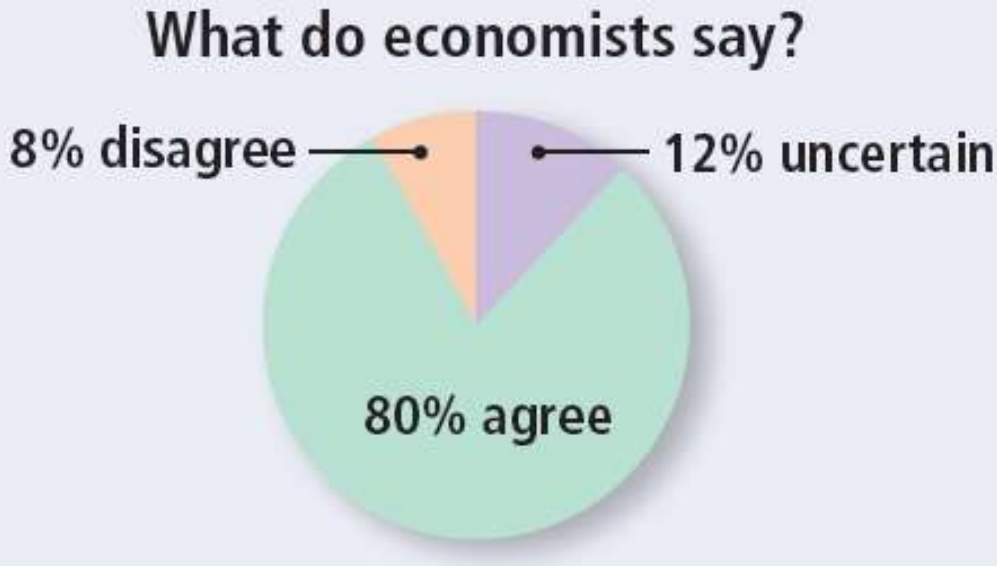
Why Economists Disagree

- Economists often give conflicting policy advice 對實證理論正確性感到懷疑
  - Can disagree about the validity of alternative positive theories about the world
  - May have different values and, therefore, different normative views about what policy should try to accomplish 價值觀不同
- Yet, there are many propositions about which most economists agree

ASK THE EXPERTS

Ticket Resale

"Laws that limit the resale of tickets for entertainment and sports events make potential audience members for those events worse off on average."



Propositions about Which Most Economists Agree (and percentage of economists who agree)

- A ceiling on rents reduces the quantity and quality of housing available. (93%) 租金天花板
- Tariffs and import quotas usually reduce general economic welfare. (93%) 進口配額
- The United States should not restrict employers from outsourcing work to foreign countries. (90%) 外包
- The United States should eliminate agricultural subsidies. (85%) 農業補貼
- Local and state governments should eliminate subsidies to professional sports franchises (85%) 專業運動機構補貼(e.g., NBA, MLB)

Propositions about Which Most Economists Agree (and percentage of economists who agree)

現金比實物支付好

- Cash payments increase the welfare of recipients to a greater degree than do transfers-in kind of equal cash value. (84%)
- A large federal budget deficit has an adverse effect on the economy. (83%) 聯邦預算赤字
- The United States should not ban genetically modified crops. (82%) 基因改造農作物
- A minimum wage increases unemployment among young and unskilled workers. (79%)
- Government subsidies on ethanol in the United States should be reduced or eliminated. (78%)

補貼生質燃料

Appendix Question

- Classify each of the following statement as positive or normative
  - Society faces a short-run trade-off between inflation and unemployment
  - A reduction in the rate of money growth will reduce the rate of inflation
  - The Federal Reserve (美國聯邦儲備委員會) should reduce the rate of money growth (貨幣增長率)
  - Society ought to require welfare recipients to look for jobs
  - Lower tax rates encourage more work and more saving

Appendix

Who Makes the Dreamliner?



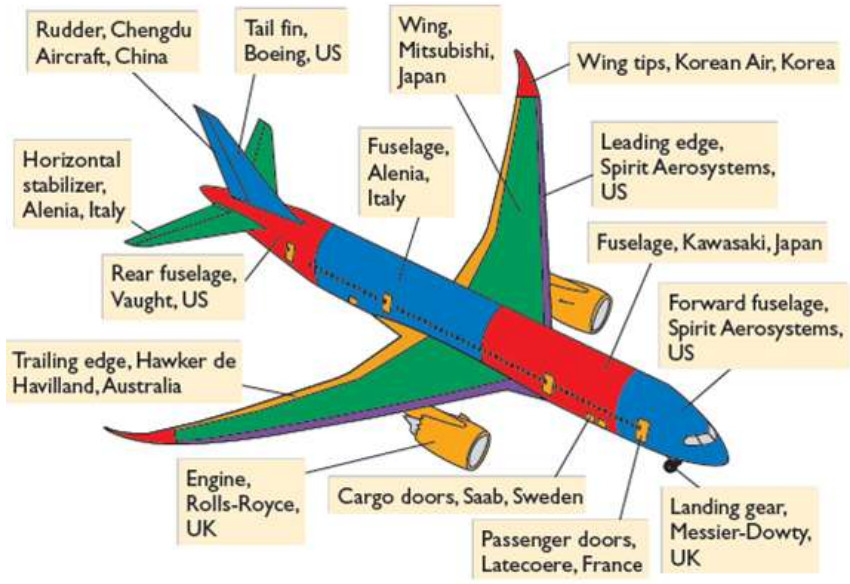
Appendix

Eye on the Dreamliner (1 of 2)

Who Makes the Dreamliner?

Boeing wants to manufacture the Dreamliner at the lowest possible cost.

Boeing achieves this goal by having more than 400 firms on 4 continents make its components.

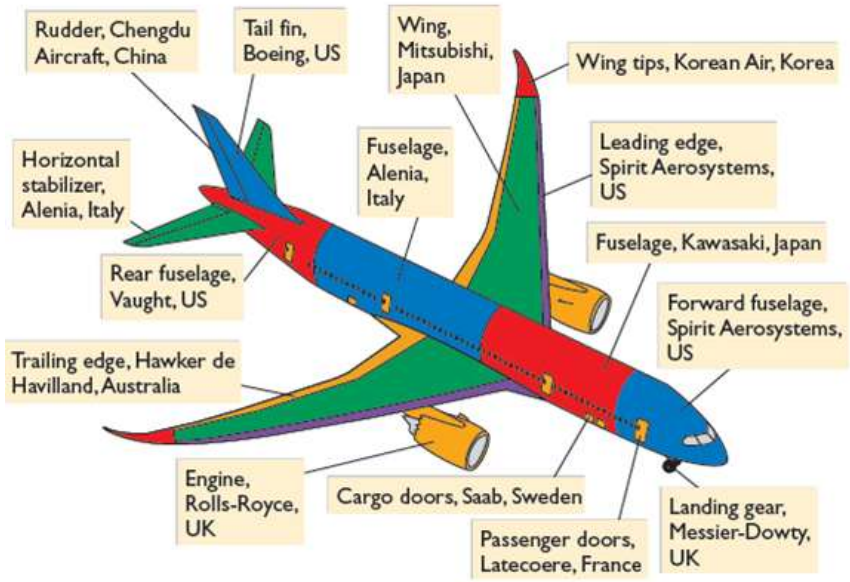


Appendix

Eye on the Dreamliner (2 of 2)

Who Makes the Dreamliner?

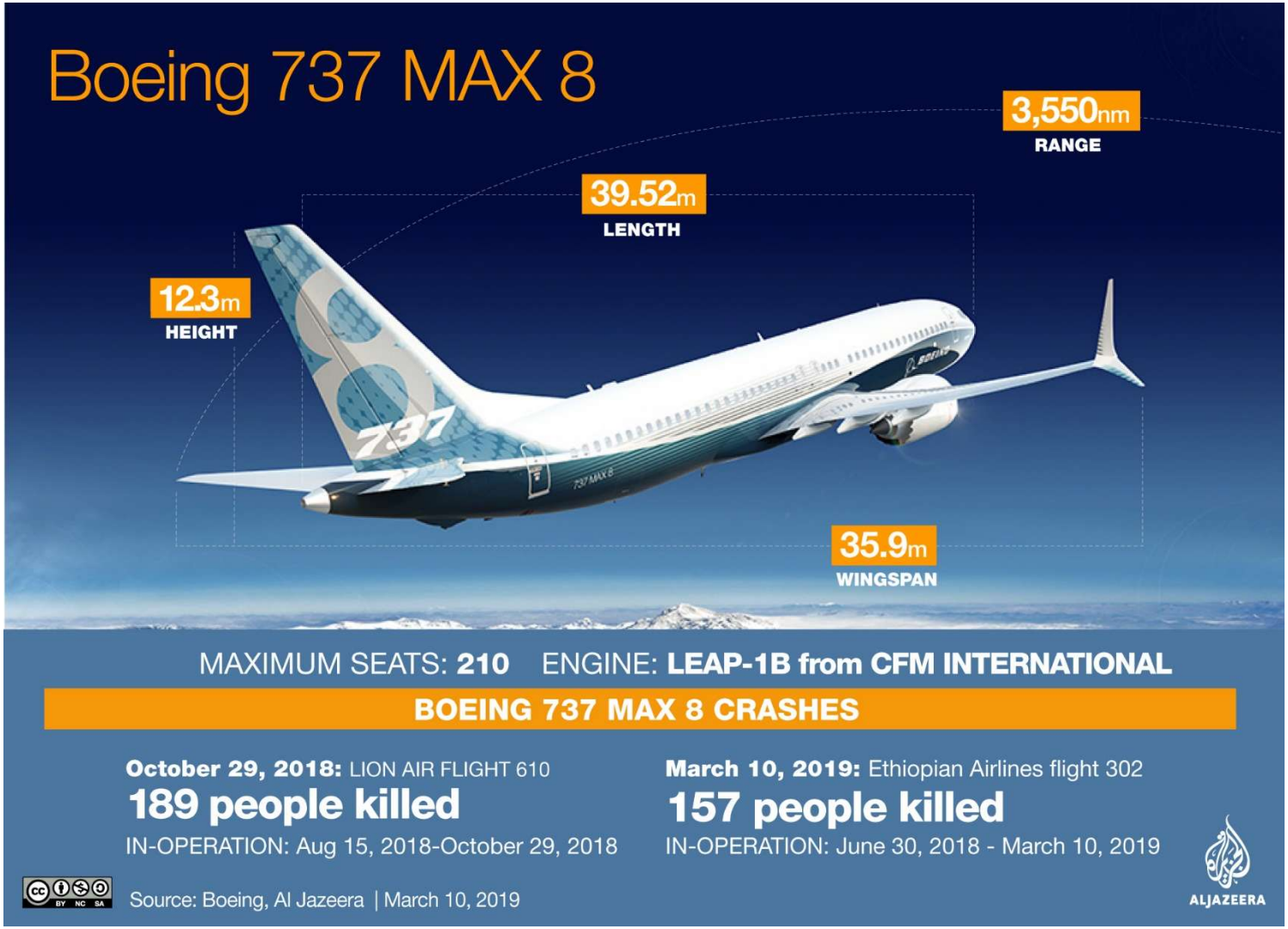
Boeing and the 400-plus firms make decisions and pay their workers, investors, and raw material suppliers to play their parts in influencing *what, how,* and *for whom* goods and services are produced.



All these decisions are made in self-interest and the Dreamliner is produced at the lowest possible cost.

Appendix

Eye on the Nightmare



Appendix

Is Wind Power Free?



Appendix

Eye on the Environment (1 of 2)

Is Wind Power Free?

Wind power is not free.

Its opportunity cost includes the huge amounts of other goods and services we give up to build wind turbines and transmission lines.

Wind turbines can produce electricity only when the wind is strong enough, but advances in technology are enabling turbines to operate in lower wind conditions.

Also some of the best wind farm locations are a long way from major population centers, so transmission lines would be long and power transmission losses large.

Appendix

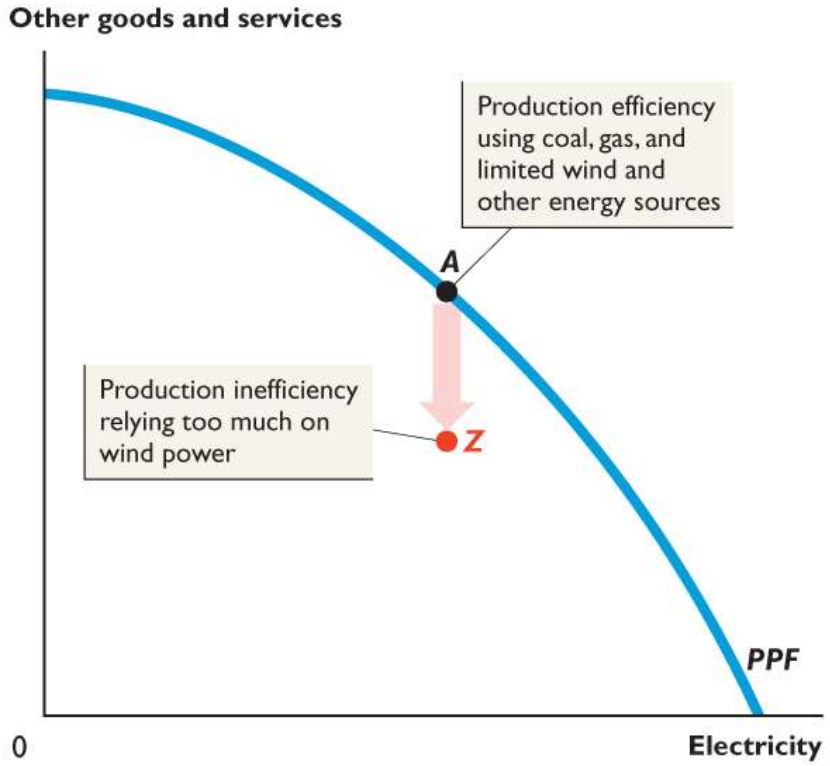
Eye on the Environment (2 of 2)

Is Wind Power Free?

Point *A* is a point of efficient electricity production.

If the United States produces most of its electricity using wind power, it would be operating inside the *PPF* at a point such as *Z*.

Point Z is a point of inefficient electricity production.



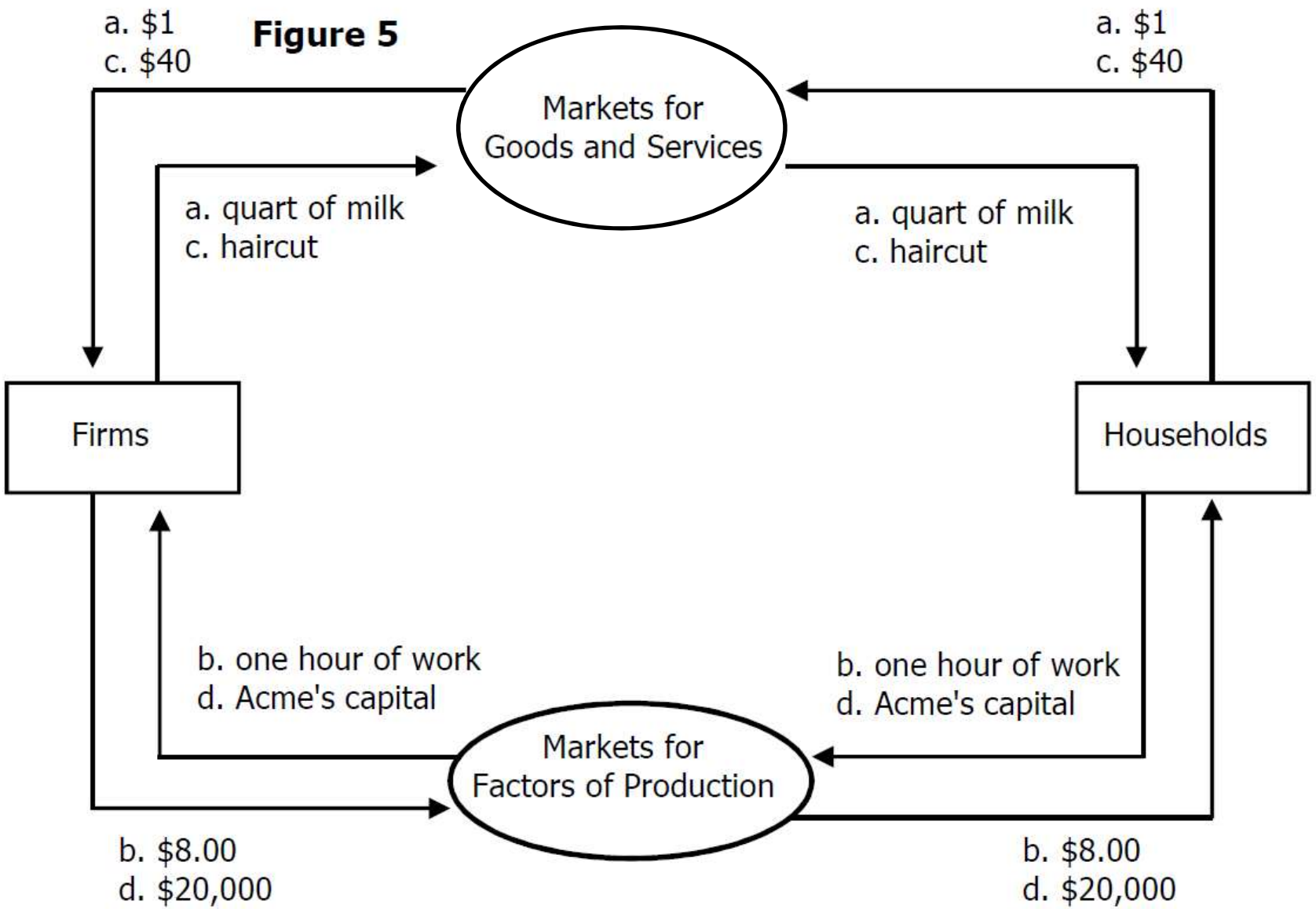
Summary

- Economists are scientists
  - Make appropriate assumptions and build simplified models
  - The circular-flow diagram and the production possibilities frontier
- Microeconomists study decision making by households and firms and their interactions in the marketplace
- Macroeconomists study the forces and trends that affect the economy as a whole

Summary

- A positive statement is an assertion about how the world is
- A normative statement is an assertion about how the world ought to be
- As policy advisers, economists make normative statements
- Economists sometimes offer conflicting advice
  - Differences in scientific judgments
  - Differences in values

Solution


Figure 5
a. $1
c. $40
a. $1
c. $40
Markets for
Goods and Services
a. quart of milk
c. haircut
a. quart of milk
c. haircut
Firms
Households
b. one hour of work
d. Acme's capital
b. one hour of work
d. Acme's capital
Markets for
Factors of Production
b. $8.00
d. $20,000
b. $8.00
d. $20,000